

Utilization of Risk Stratification in Patients with Diabetes Mellitus in a Health System Specialty Pharmacy Setting

Mackenzie Stout, PharmD; Nicholas McDonald, PharmD; Lauren Bryant, PharmD, CDCES; Casey Ross, PharmD, CTTS, MBA; Ana Lopez Medina, PharmD; Carly Giavatto, PharmD; Jessica Mourani, PharmD

BACKGROUND

- Diabetes has a high economic burden, accounting for one in four healthcare dollars ¹
- It is estimated to cost over \$16,000 on average to care for a patient with diabetes annually ¹
- Integration of pharmacists into health systems result in significant hemoglobin A1C (A1C) reductions and improved patient outcomes in patients with diabetes mellitus ²
- Limited research has been conducted illustrating how to adopt chronic disease management (CDM) services into a health system specialty pharmacy (HSSP) workflow
- Risk stratification has been proven successful in other disease states in prioritizing high risk patients ³

OBJECTIVES

The purpose of this study is to assess the effect of patient outcomes after implementing a risk stratification protocol in the HSSP setting for patients with diabetes

METHODS

Study Design

This is a retrospective descriptive study of adult patients enrolled in the CDM program from July 2020 to August 2022

Patients were risk stratified based on A1C value

Table 1. Risk Stratification Criteria

High Risk (n=88)	Low Risk (n=372)	Not Enrolled (n=100)
- A1C ≥ 10 - Received proactive high-touch clinical education and support from pharmacists - Moved to low-risk when A1C < 8	- A1C < 10 - Received personalized clinical support and education from HSSP staff	- No clinical support or education from pharmacists - Did not fill prescriptions with health system specialty pharmacy

Setting

Southern Ohio Medical Center's (SOMC) outpatient family practice offices

Inclusion and Exclusion Criteria

Inclusion: Diagnosis of diabetes, filled diabetes medication with SOMC's community pharmacy, and had at least two A1C values recorded

Exclusion: Pediatric patients, prescribed diabetes medications for other conditions

Endpoints

- Average change of A1C from baseline to most recent follow-up A1C in which at least three months had elapsed
- Average time for patients to move from high to low risk group
- Pharmacist intervention reasons and acceptance rates of those interventions

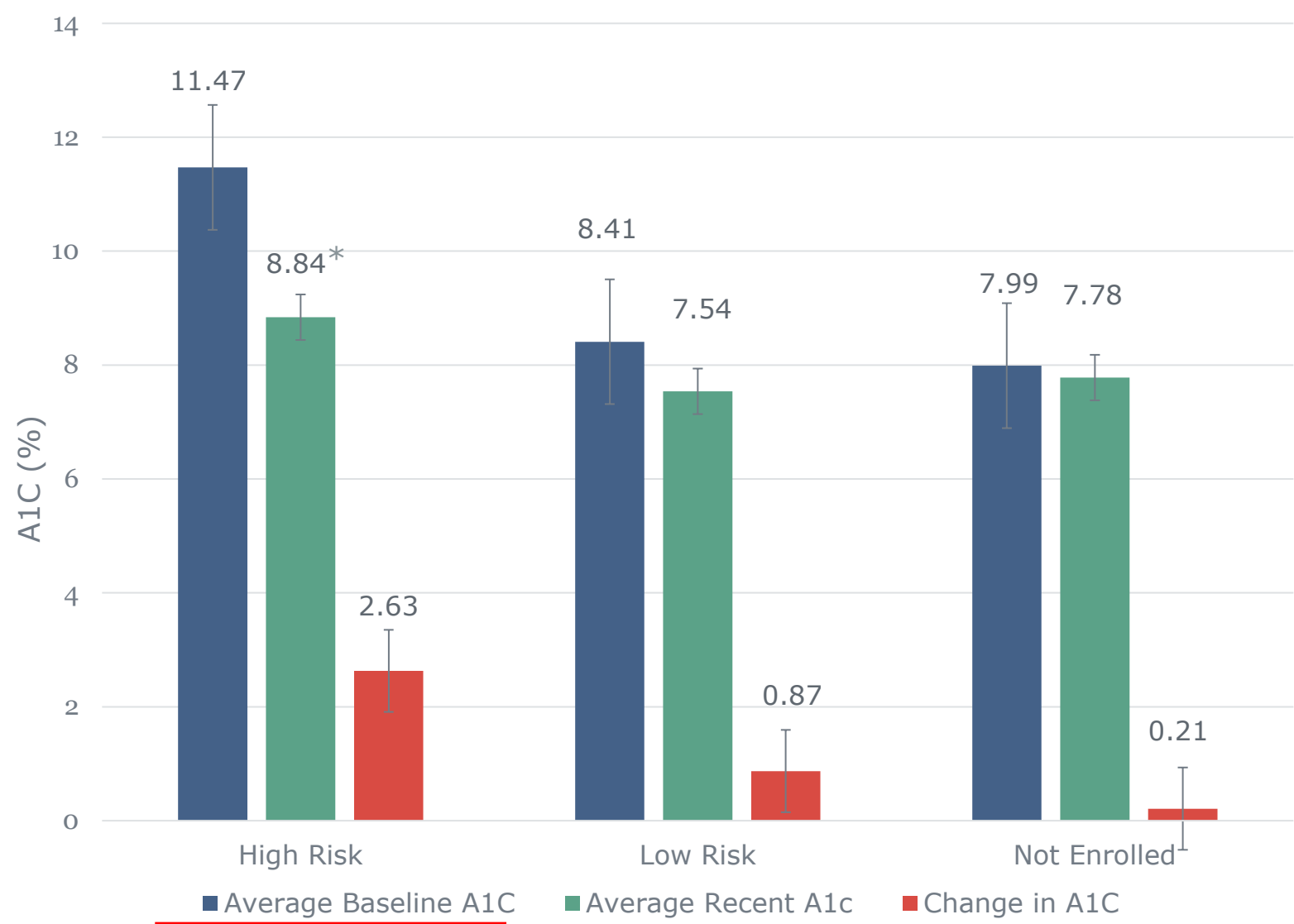
Statistical Analysis

- Average baseline and recent A1C was compared using a Paired t-Test
- P-value < 0.05 was considered statistically significant
- All statistical analysis were completed in R Programing Software

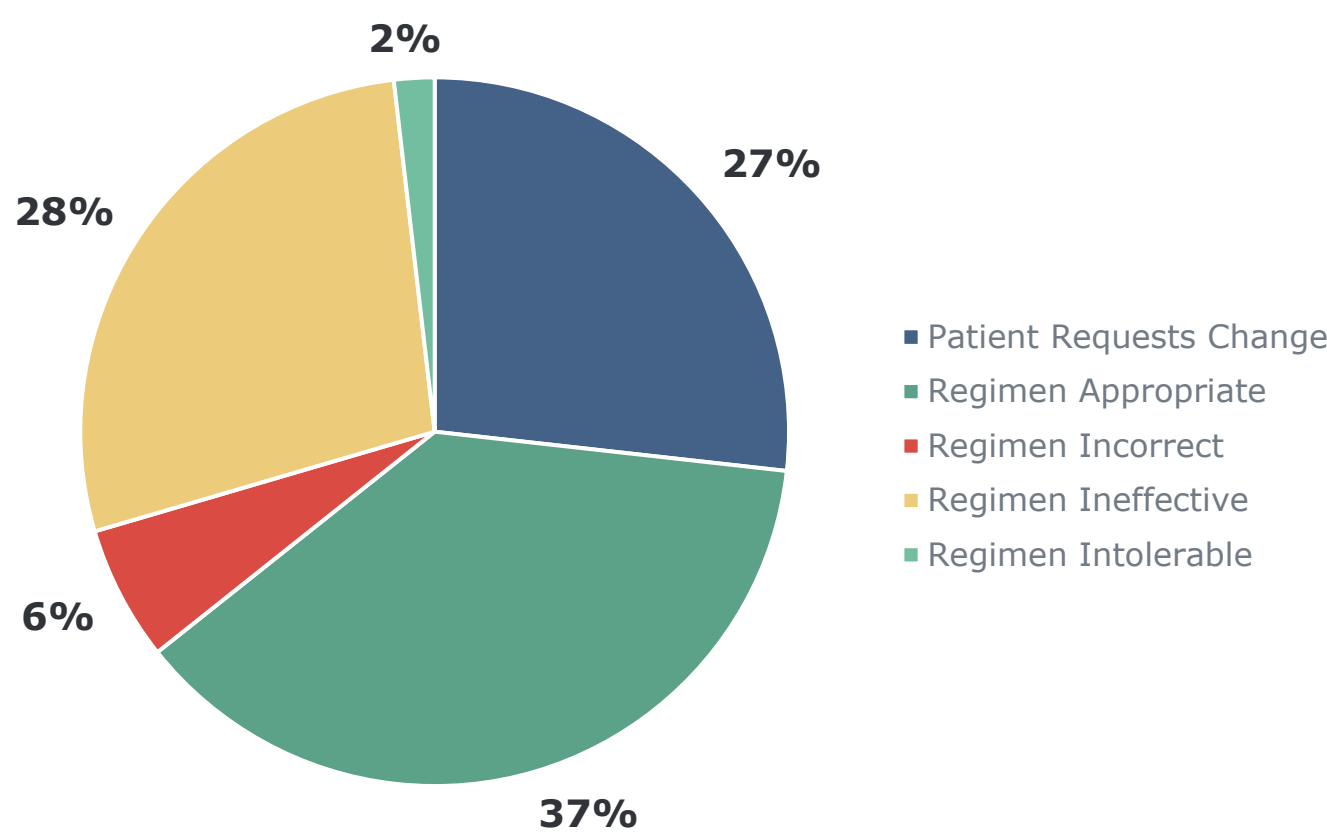
RESULTS

A1C Reduction by Risk Stratification

High risk patients saw an average A1C change of 23%



Regimen Intervention Reason (n=325)

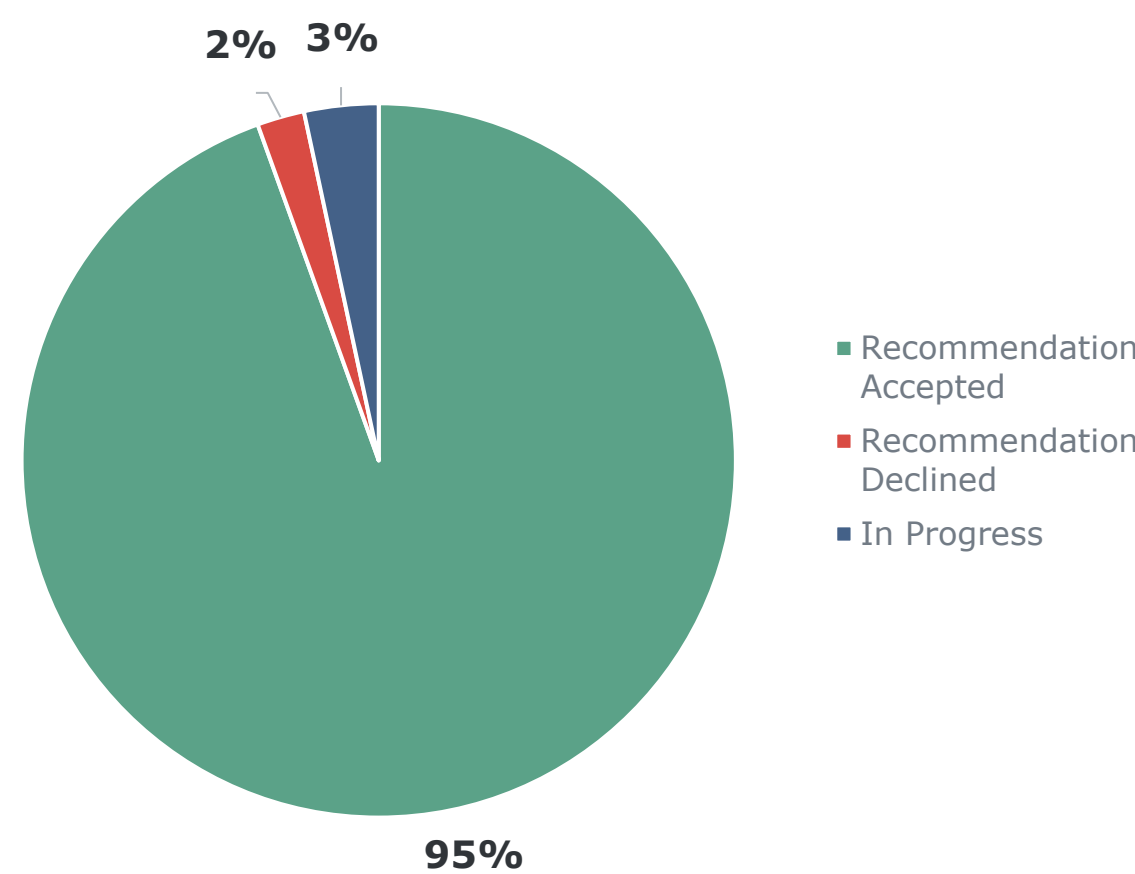


Risk Improvement

71% of high-risk patients moved to low-risk with HSSP CDM program

Number of Patients Moved to Low Risk	63
Average Baseline A1C	10.97
Average Recent A1C	7.04
Change in A1C	3.93
Average Days to Low Risk	192

Intervention Outcome



COST SAVINGS

Reduction in Diabetes Healthcare Cost for High Risk Patients*	Health System Financial Assistance	Manufacturer Copay Assistance
Per Patient	\$1,936	
All High Risk Patients	\$170,368	\$84,653

*Reductions in A1C by 1% may reduce annual diabetes related healthcare costs per patient \$736 ⁴

CONCLUSIONS

A1C Reduction

- Patients stratified into both the high- and low-risk groups had greater reductions in A1C compared to patients who did not use CDM services
- High Risk Patients saw an average A1C change of 23% from baseline
- Low risk patients decreased A1Cs, and remained controlled, demonstrating that efficacy of risk stratification

Risk Stratification

- The risk stratification model allows high volume chronic diseases like diabetes mellitus to be operationalized by specialty pharmacy.
- Risk stratification in DM results in both best patient outcomes and profit by allocation of pharmacist resources and 340b cost savings

Pharmacist Intervention

- Pharmacists successfully identified regimen interventions in between provider visits to improve regimen efficacy
- 63% of these interventions resulted in regimen changes
- Interventions had a 95% acceptance rate by both patients and providers

Impact of HSSP

- This study demonstrates the impact HSSP pharmacists have on diabetes management and their ability to manage this high volume patient population
- It also outlines the benefits of a risk stratification protocol in patients with diabetes

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